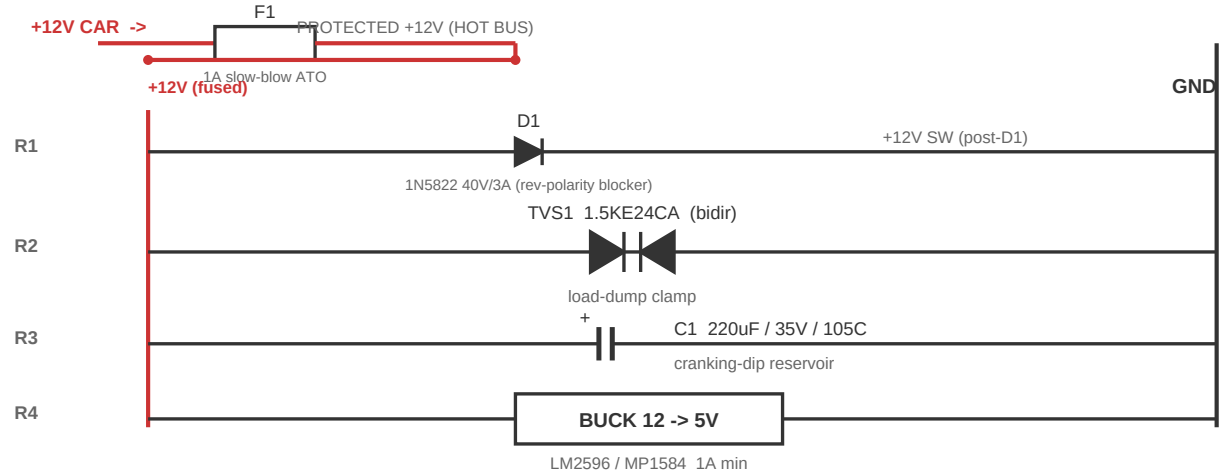


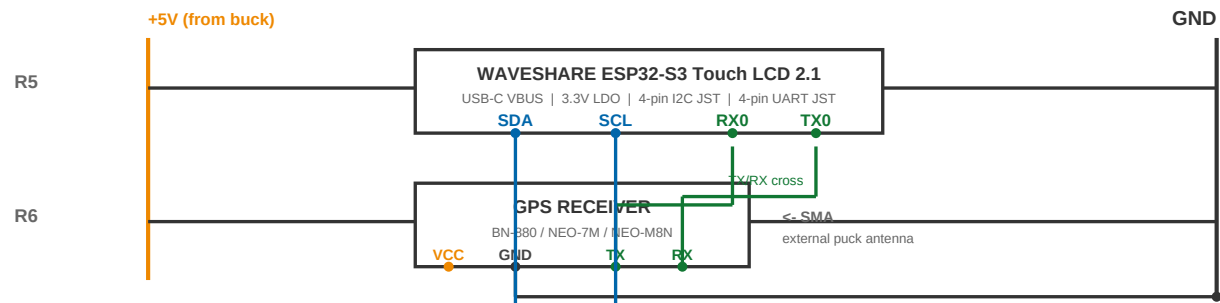
FIREBIRD MULTI-GAUGE - Ladder Diagram (Rev 5)

+12V POWER INPUT PROTECTION

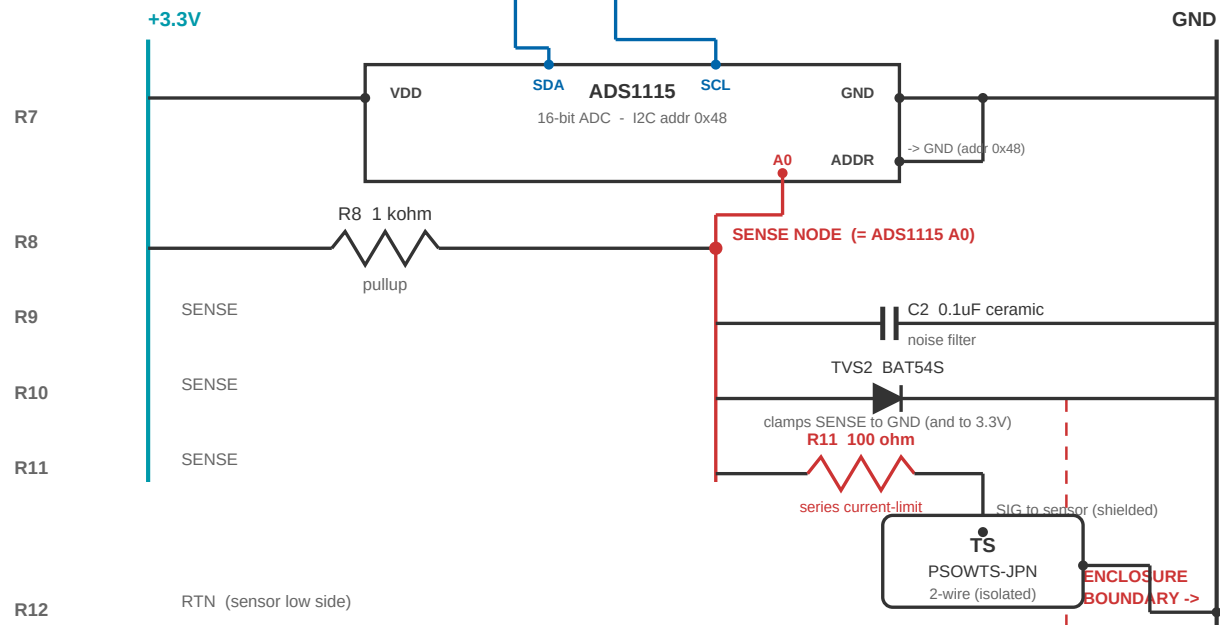


BENCH BUILD R1: jumper D1 until 1N5822 arrives. R2: leave TVS1 pads empty until 1.5KE24CA arrives.

+5V -> WAVESHARE + GPS



SENSOR INPUT PROTECTION (3.3V from Waveshare I2C port)



Rev 5 - Legend, BOM, Changelog

LEGEND - schematic symbols

	Resistor Carbon or metal-film resistor (R8, R11)
	Diode (Schottky) Reverse-polarity blocker (D1 = 1N5822)
	TVS, bidirectional Two zeners back-to-back; clamps over-voltage either polarity (TVS1)
	Capacitor, electrolytic Polarized; bulk reservoir (C1 = 220uF)
	Capacitor, ceramic Non-polarized; noise filter (C2 = 0.1uF)
	Junction (connection) Filled dot = wires connected
	Fuse F1 = 1A slow-blow ATO (input fuse, protects entire 12V circuit)
	Sensor, 2-wire isolated TS body floats; both leads return to board

BILL OF MATERIALS

F1	Fuse 1A slow-blow ATO + inline holder (input)
D1	1N5822 Schottky, 40V / 3A axial
TVS1	1.5KE24CA bidirectional (load-dump clamp)
C1	220uF 35V 105C electrolytic, radial
TVS2	BAT54S SOT-23 OR 2x 1N5817 axial
C2	0.1uF 50V ceramic disc
R8	1 kohm 1% 1/4W metal-film (pullup)
R11	100 ohm 1% 1/4W metal-film (limit)
BUCK	12V -> 5V buck (LM2596 / MP1584)
ADS1115	16-bit I2C ADC, addr 0x48 (ADDR=GND)
ESP32-S3	Waveshare ESP32-S3 Touch LCD 2.1
GPS	BN-880 / NEO-7M / NEO-M8N (5V) + SMA puck antenna
TS	Prosport PSOWTS-JPN 2-wire coolant temp sender

CHANGELOG

Rev 2	TVS2 changed from P6KE6.8A to BAT54S / 1N5817. P6KE6.8 clamps ~10.5V, too high for ADS1115 (3.6V abs max).
Rev 2.1	Sender corrected to Prosport PSOWTS-JPN. BENCH notes added for what to do until D1 / TVS1 arrive.
Rev 3	Ladder redrawn: single rung labels, BENCH notes in callout, changelog broken out of BOM.
Rev 4	Treat sender as 2-wire isolated. ADS1115 pinout shown explicitly. Bidirectional TVS1 symbol redrawn correctly. Crossing-wire convention noted. Legend page added.
Rev 5	F1 MOVED: now in series with the +12V input lead BEFORE the rail split. Previously F1 was inside D1's branch only -- TVS1, C1 and BUCK all tapped from an unfused rail. Rev 5 puts F1 in the input so every downstream branch is fuse-protected. GPS RECEIVER ADDED: BN-880/NEO-7M/NEO-M8N module on the +5V rail (R6), parallel to the Waveshare. UART data lines (TX/RX) cross-wired to the Waveshare's UART JST. External SMA puck antenna noted at the GPS. Rung references renumbered (former R5 -> still R5; ADS1115 R6->R7, pullup R7->R8, C2 R8->R9, TVS2 R9->R10, R10->R11, RTN R11->R12).

STAR-GROUND: all GND connections meet at one point on the protoboard. Keep the long sensor wire shielded; route away from ignition coils.